



SAGAR PUBLIC SCHOOL, BHOPAL
HOLIDAY HOMEWORK, 2025-26
CLASS: IX

PHYSICS (CH.-7, MOTION)

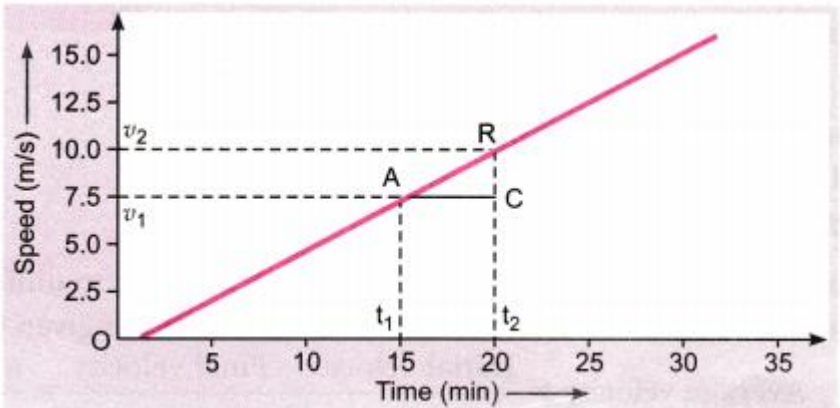
General Instructions:

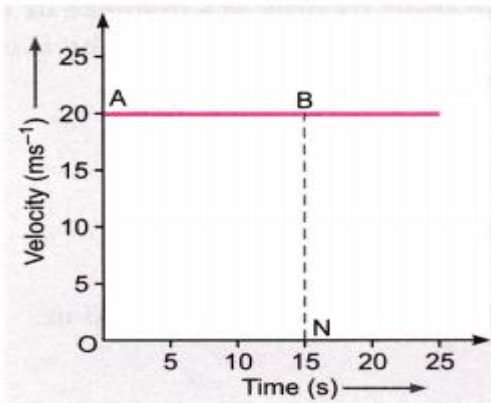
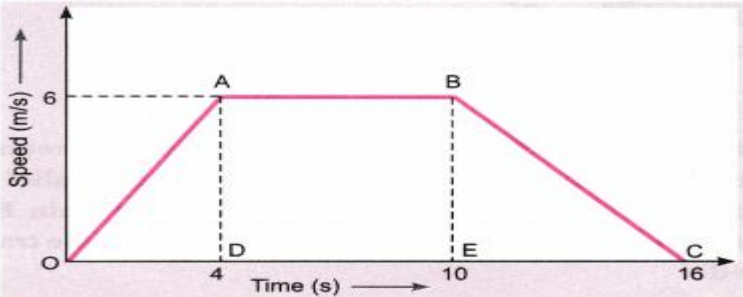
Total no. of printed pages: 15

● TO BE DONE IN CLASS WORK NOTEBOOK

Q.1.	The direction of acceleration is the same as the direction of _____. (a) displacement (b) change in velocity (c) velocity (d) all of these	
Q.2.	The area under a velocity-time graph gives _____. (a) distance (b) acceleration (c) speed (d) displacement	
Q.3.	_____ can be added algebraically. (a) Accelerations (b) Velocities (c) Distances (d) Any physical quantities	
Q.4.	The distance-time graph of a body moving along a straight path in a single direction with uniform speed will be _____. (a) along X-axis (b) a line with positive slope (c) parallel to X-axis (d) None of these	
Q.5.	A physical quantity which has only magnitude is called a _____. (a) scalar quantity (b) vector quantity (c) neither (a) nor (b) (d) either (a) or (b)	
Q.6.	The position-time graph of a body is parabolic, then the body is _____. (a) in non-uniform motion (b) in uniform acceleration (c) at rest with some initial distance covered (d) at rest with zero initial distance covered	
Q.7.	The slope of a velocity-time graph gives _____. (a) distance (b) speed (c) acceleration (d) displacement	

Q.8.	A body travels along a straight path from its initial position to a point 20 m away and then returns back to its initial position. The change in the position of the body is ____ and the distance travelled is ____. (a) 20 m, 40 m (b) 0 m, 40 m (c) 40 m, 40 m (d) 0 m, 0 m	
Q.9.	A freely falling body is said to be moving with ____. (a) constant non-zero acceleration (b) non-uniform motion (c) zero velocity (d) non-uniform acceleration	
Q.10.	If a body travels unequal distances in equal intervals of time along a _____ path, the body is said to be in _____. (a) curved, uniform motion (b) pentagonal, uniform motion (c) rectangular, uniform motion (d) straight, non-uniform motion	
	<u>Directions:</u> In each of the following questions, a statement of Assertion is given and a corresponding statement of Reason is given just below it. Of the statements, given below, mark the correct answer as: (a) Both assertion and reason are true and reason is the correct explanation of assertion. (b) Both assertion and reason are true but reason is not the correct explanation of assertion. (c) Assertion is true but reason is false. (d) Both Assertion and Reason are false.	
Q.11.	Assertion: An object may acquire acceleration even if it is moving at a constant speed. Reason: With change in the direction of motion, an object can acquire acceleration.	
Q.12.	Assertion: The graph between two physical quantities P and Q is straight line, when P/Q is constant. Reason: The straight line graph means that P is proportional to Q or P is equal to constant multiplied by Q.	
Q.13.	Assertion: An object may have acceleration even if it is moving with uniform speed. Reason: An object may be moving with uniform speed but it may be changing its direction of motion.	
Q.14.	Assertion: The displacement of an object can be either positive, negative or zero. Reason: Displacement has both the magnitude and direction.	
Q.15.	Assertion: When the displacement of a body is directly proportional to the square of the time. Then the body is moving with uniform acceleration. Reason: The slope of velocity-time graph with time axis gives acceleration.	

Q.16.	A particle is moving in a circular path of radius r . What would be the displacement after half a circle?	
Q.17.	Give an example of a body which may appear to be moving for one person and stationary for the other.	
Q.18.	(a) Name the physical quantity measured by the area occupied below velocity-time graph for uniform motion. (b) The displacement of a moving object in a given interval of time is zero. Would the distance travelled by the object also be zero? Justify your answer.	
Q.19.	(a) Name the physical quantity that corresponds to the rate of change of velocity. (b) What do you mean by negative acceleration? Give one example of retardation or negative acceleration.	
Q.20.	(a) If the displacement-time graph for a particle is parallel to time axis, how much is the velocity of the particle. (b) What will you say about the motion of a body if its distance-time graph is a straight line having a constant angle with time axis? (c) What does the slope of a velocity-time graph represent?	
Q.21.	(a) Distinguish between distance and displacement. (b) Mention two point Of difference between speed and velocity. (c) What is the difference between uniform motion and non-uniform motion?	
Q.22.	Given below is the velocity-time graph for the motion of the car. What does the nature of the graph show? Also find the acceleration of the car. 	

Q.23.	The velocity-time graph shows the motion of a cyclist. Find (i) its acceleration (ii) its velocity and (iii) the distance covered by the cyclist in 15 seconds. 
Q.24.	Study the speed-time graph of a body given here and answer the following questions:  (a) What type of motion is represented by OA? (b) What type of motion is represented by AB? (c) What type of motion is represented by BC? (d) Find out the acceleration of the body. (e) Calculate the retardation of the body. (f) Find out the distance travelled by the body from A to B.
	Art Integrated Activity (Roll No. 1 to 13) Prepare a PPT on Improvement in agricultural practices comparing states of Bihar and Madhya Pradesh using technology.

Holiday Home Work Cl. 9 Chemistry 2025-26

1. Concept Understanding

Q:1 What are the characteristics of particles of matter? Explain each with one daily life example.

Q2. Differentiate between solids, liquids, and gases on the basis of:

- Shape
- Volume
- Compressibility
- Fluidity
- Inter-particle space

Q3. What happens at the particle level when a solid melts?

2. Application-Based Questions

Q:1 Investigate why it takes longer to cook food in hilly areas and explain how pressure affects the boiling point of water. Make a diagram to explain it.

Q:2 Place cold water in a **matka** and another in a steel bottle. Observe the temperature difference after 2 hours.

Explain:

- Why does the matka water remain cooler?
- Which process is involved?

3. Activity-Based Learning

Perform a simple home experiment: Take two bowls — one with water and another with water + sugar. Keep both under the fan. Record which evaporates faster and why.

4. Art Integration (Roll No. 14 to 26)

Comic Strip: "A Water Molecule's Journey"

Task:

Prepare a comic strip ppt showing a water molecule's journey from:

- Ice → Water → Steam → Cloud → Rain
Include Bihari culture scenes like handis (earthen pots), monsoon in villages, and pujas.

OR

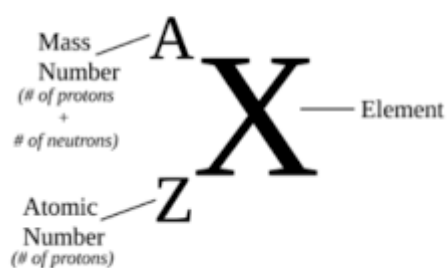
Creative Cards with 3D Symbols (Bihar Handicraft Style)

Task:

Design colorful 4x4 inch cards for 4 elements (based on your roll number) using:

- Madhubani Art
- Add: symbol, atomic number, atomic mass for example

Atomic Symbol



CLASS IX BIOLOGY Holiday Homework



Q.1. Observe the picture and answer the following questions:

- Identify the disease of wheat crop.
- Write its causes.
- How can it be prevented?

Q.2. a. What is Biofortification ?

b. Good as Gold: Can Golden Rice and Other Biofortified Crops Prevent Malnutrition? What is golden rice and how is it good for our health?

Q.3. Eutrophication is a leading cause of impairment of many freshwater and coastal marine ecosystems in the world. Why should we worry about eutrophication and how is this problem managed?

<https://oceanservice.noaa.gov/facts/eutrophication.html>

Q.4. Read the chapter Improvement in food resources and answer the questions below.

- Why is India an agriculturally important country?
- What are the salient features of Indian agriculture?
- Explain plantation agriculture.

Q.5. Find out the cooperative societies which are helping farmers for marketing of their produce. (any from your state or any other state of India).

Q6. Make your own liquid Fertilizer. Write its advantages. Click the picture and paste it in your notebook.

Art integration Project (Roll No. 27 - 40)

“Natural dye yielding plants and indigenous knowledge of dyeing in Madhya Pradesh and Bihar ”

Select any natural dye from Bihar or Madhya Pradesh and make a drop down card on the steps of making natural dye along with a block printed decorative card by using the selected dye.

Note : Q 1 to Q 6 to be done in the notebook

HINDI (ग्रीष्मकालीन अवकाश गृहकार्य)

रोल नंबर (1-40) प्रश्न 1 'दो बैलों की कथा' पाठ की विधा, उद्देश्य, पात्र परिचय, शिक्षा, मानचित्र (माइन्ड मैप)

प्रश्न 2 पिंजरबद्ध और आकाश में उन्मुक्त उड़ान भरने वाले दो पक्षियों के मध्य होने वाले वार्तालाप को संवाद-शैली में लिखते हुए स्पष्ट करें कि स्वतंत्रता का जीवन में क्या महत्त्व है?

प्रश्न 3

रोल नंबर (1-20) बिहार और मध्यप्रदेश के जनजीवन, भाषाओं तथा हस्तशिल्प पर पीपीटी / वीडियो

रोल नंबर (21-40) बिहार और मध्यप्रदेश के साहित्यकारों का परिचय (पी पी टी/ वीडियो)

Computer Holiday Homework, 2025-26

1. Webpage

Software – HTML

Prepare a webpage to show comparison between Madhya Pradesh and Bihar States including following points:

- Cultural Heritage
- Food
- Literacy Rate
- Images of famous places
- Send webpage to your Subject Teacher on mail.
 - a. artisaxena@spsbhopal.ac.in
 - b. chayanpreetwalia@spsbhopal.ac.in
 - c. vinodkumarsingh@spsbhopal.ac.in

and

2. Write answer of following questions in the IT notebook:

Q1: What are the advantages and disadvantages of verbal communication?

Q2: Explain 7C's of communication.

Q3: Give an example for the following:

- Clear communication
- Complete communication.

Q4: Write down the seven factors affecting perspectives in communication.

Q5: Explain five types of non-verbal communication.

Q6: Write one sentence for each type of sentences:

- Statement
- Question
- Exclamatory
- Order

SAKET NAGAR

SOCIAL SCIENCE HOLIDAY HOMEWORK CLASS-IX

1. Project Work on Disaster Management (To be assessed as Multiple Assessment as CBSE curriculum 2025-26)

Overview

This project work is designed to help students understand the various facets of disaster management. The project focuses on creating disaster awareness, evaluating preparedness measures, and developing strategies for effective management. Students will work collaboratively using research, surveys, interviews, art integration, and creative solutions to explore the topic.

Research Questions: Select one of the Disaster and prepare project file work on following parameters.

Impact Analysis: How do natural and man-made disasters affect the social, economic, and environmental aspects of communities?

- Investigate short-term and long-term effects.
- Identify vulnerable groups and economic repercussions.
- Use case studies to illustrate these impacts.

Community Preparedness: What measures can local communities adopt to prepare for disasters, and how effective are these strategies?

- Assess the role of early warning systems, evacuation plans, and emergency drills.
- Evaluate the contributions of local disaster management authorities.
- Suggest improvements based on local needs.

Government and Institutional Role: How do government schemes and institutional mechanisms contribute to disaster management, and what are the key challenges in their implementation?

- Examine specific government initiatives for disaster risk reduction.
- Analyze the collaboration between governmental bodies and local organizations.
- Propose recommendations to bridge gaps in policy and practice.

Phases of Disaster Management: What are the key phases of disaster management (preparedness, response, recovery, and mitigation), and how can each phase be optimized for better outcomes?

- Define each phase with real-world examples.
- Identify challenges and best practices in each stage.

- Recommend comprehensive strategies to strengthen overall management.

Subtopics to be Covered

- 1. Introduction to Disaster Management:** Definition and importance
Overview of different types of disasters (natural vs. man-made)
- 2. Impact of Disasters:** Social, economic, and environmental consequences
Case studies of significant disasters
- 3. Preparedness Measures:** Early warning systems and evacuation plans
Role of education and community drills
- 4. Response and Recovery:** Emergency response protocols
Strategies for recovery and rehabilitation
Analysis of government schemes and relief measures
- 5. Mitigation Strategies:** Long-term planning for disaster risk reduction
Role of technology and innovative solutions
Integration of art and community participation in awareness campaigns

NOTE- Student should be allowed to submit the file in handwritten or digital format as per the CBSE curriculum 2025-26

2. Interdisciplinary Project

(To be assessed as Subject Enrichment Activity as per CBSE Curriculum 2025-26)

Topic- Forest, Society and Colonialism

OBJECTIVE- This project is designed to integrate history, geography, political science, and economics by exploring how forests and societies have interacted through different periods. Use inquiry-based learning, cooperative discussions, research, and art integration to answer the questions and understand the complexities of forest management in various eras.

Project Questions- (Group Activity)

- 1. Evolution of Forest Societies (Roll no- 01 to 05 , 11 to 15)**
 - How did traditional practices and forest management evolve from the pre-colonial era to the colonial era?
 - *Guideline:* Investigate indigenous practices, cultural beliefs, and local approaches to forest conservation.
- 2. Impact of Colonial Policies on Forests (Roll no- 06 to 10, 16 to 20)**
 - In what ways did colonial policies alter the structure and function of forests, and how did these changes pave the way for commercial forestry?
 - *Guideline:* Compare pre-colonial traditional methods with the systematic exploitation introduced during colonialism.
- 3. Rebellions and Resistance (Roll No- 21 to 25 , 31 to 35)**
 - What were the key reasons behind forest-based rebellions in South-East Asia, particularly in Java, and how did these revolts reflect the responses to colonial exploitation?
 - *Guideline:* Examine historical episodes, using case studies such as Java, to understand the triggers and outcomes of these rebellions.

4. Role of Government and Community in Conservation (Roll no- 26 to 30, 36- 40+)

- How did local communities and government interventions work together to conserve forest cover in different historical periods?
- *Guideline:* Analyze initiatives, policies, and community actions that contributed to the safeguarding of natural resources over time.

Subtopics to be Covered

- **Pre-Colonial Era**
 - Traditional forest management and indigenous conservation practices
 - Cultural beliefs and sustainable use of natural resources
- **Colonial Impact**
 - Colonial policies on land and forest management
 - Introduction and expansion of commercial forestry
 - Changes in societal structure and exploitation of natural resources
- **Forest-Based Rebellions**
 - Case study focus: Rebellions in South-East Asia with an emphasis on Java
 - Causes and triggers of resistance against enforced forestry policies
- **Post-Colonial and Conservation Efforts**
 - Government actions in forest conservation after colonial rule
 - Role of local communities and collaborative initiatives
 - Evaluation of modern forestry practices influenced by historical context

Note- Students should prepare a comprehensive Power point Presentation and should include evidences while presenting the topic, if any.

*अवकाश कालीन गृहकार्य कक्षा नवमी- SANSKRIT

सभी के लिए - 1-अस्मद्, युष्मद् के शब्दरूप लिखिए।

2- मध्यप्रदेश की प्राचीन शिक्षा पद्धति पर चार्ट बनाइए।

रोल न. 1-20 उपपद विभक्ति (द्वितीया, तृतीया एवं चतुर्थी) प्रत्येक के 10-10 वाक्य

रोल न.21-40 उपपद विभक्ति (पंचमी, षष्ठी एवं सप्तमी) प्रत्येक के 10-10 वाक्य

Maths Holiday Homework

CLASS: IX

Section – A

(Subjective Questions)

Q 1. Express the following decimals in the form $\frac{p}{q}$.

(i) $0.\overline{32}$ (ii) $0.12\overline{3}$ (iii) $0.003\overline{52}$ (iv) $15.7\overline{12}$

Q 2. Find a rational number between $-\frac{2}{3}$ and $\frac{1}{4}$.

Q 3. Evaluate: $2^{55} \div 2^{60} - 2^{97} \div 2^{18}$.

Q 4. If $a = 2$ and $b = 3$, then find the values of each of the following :

(i) $a^a + b^b$ (ii) $a^b + b^a$

Q 5. Simplify: (i) $\sqrt[3]{(343)^{-2}}$ (ii) $(\sqrt{x})^{2/3} \sqrt{y^4} \cdot \sqrt{xy^{-1/2}}$

Q 6. Prove that : (i) $\sqrt{3 \cdot 5^{-3}} \cdot \sqrt[3]{3^{-1}} \sqrt{5} \cdot \sqrt[6]{3 \cdot 5^4} = \frac{3}{5}$

(ii) $\sqrt{\frac{1}{4}} + (0.01)^{-1/2} - (27)^{2/3} = \frac{3}{2}$

Q 7. If $27^x = \frac{9}{3^x}$, find x .

Q 8. Find the values of x in each of the following :

(i) $2^{5x} \cdot 2^x = \sqrt[5]{2^{20}}$ (ii) $5^{x-2} \cdot 3^{2x-3} = 135$

Q 9. If the polynomials $2x^3 + ax^2 + 3x - 5$ and $x^3 + x^2 - 4x + a$ leave the same remainder when divided by $x - 2$, find the value of a .

Q 10. Simplify the following expressions :

(i) $(4 + \sqrt{7})(3 + \sqrt{2})$ (ii) $(3 + \sqrt{3})(5 - \sqrt{2})$ (iii) $(11 + \sqrt{11})(11 - \sqrt{11})$
(iv) $(5 + \sqrt{7})(5 - \sqrt{7})$ (v) $(3 + \sqrt{3})(3 + \sqrt{3})$

Q 11. Express each one of the following with rational denominator:

$$(i) \frac{2 - \sqrt{3}}{\sqrt{3}} \quad (ii) \frac{7 + 3\sqrt{5}}{7 - 3\sqrt{5}} \quad (iii) \frac{2\sqrt{3} - \sqrt{5}}{2\sqrt{2} + 3\sqrt{3}}$$

Q 12. In each of the following determine rational numbers a and b:

$$(i) \frac{\sqrt{3} - 1}{\sqrt{3} + 1} = a - b\sqrt{3} \quad (ii) \frac{4 + \sqrt{2}}{2 + \sqrt{2}} = a - b$$

Q 13. If $x = 3 + \sqrt{8}$, find the value of $x^2 + \frac{1}{x^2}$.

Q 14. If $x = 2 + \sqrt{3}$, find the value of $x^3 + \frac{1}{x^3}$.

Q15. If $x = -\frac{1}{2}$ is a zero of the polynomial $p(x) = 8x^3 - ax^2 - x + 2$, find the value of a.

Q 16. If $x + \frac{1}{x} = \sqrt{5}$, find the value of $x^2 + \frac{1}{x^2}$ and $x^4 + \frac{1}{x^4}$.

Q 17. If $x^2 + \frac{1}{x^2} = 79$, find the value of $x + \frac{1}{x}$

Q 18. If $x^2 + \frac{1}{x^2} = 66$, find the value of $x - \frac{1}{x}$.

Q 19. Factorize : (i) $x^2 + 5\sqrt{5}x + 30$ (ii) $x^2 - 2\sqrt{2}x - 30$

Q 20. Factorize each of the following algebraic expressions :

$$(i) x^8 - y^8 \quad (ii) a^{12}x^4 - a^4x^{12}$$

Section – B

(Art Integrated Project - PPT)

Applications of mathematics in art of M.P. and Bihar.